GANESH BARMAVATH

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EDUCATION

Indian Institute of Information Technology, Guwahati — Guwahati, India
B.Tech - Computer Science and Engineering — *Aug 2023 – Present*

EXPERIENCE

- **Solvrex Pvt. Limited** — **Remote**
  Software Engineering Intern — *Sep 2026-Present*
  - Contributed to development of 5+ reusable UI components for VELMORI's frontend using React.js.
  - Built and integrated 8+ RESTful API endpoints for backend functionality using Node.js and Express.js.
  - Collaborated with a 5-member team using Git/GitHub and Agile practices to deliver 15+ new features and resolve 40+ reported bugs, improving system stability by 20%.
  - Participated in code reviews and debugging sessions, reducing technical debt by 15% and improving overall code quality for 700+ concurrent users.
  - Tech Stack: React, Node.js, Express.js, HTML, CSS, JavaScript, Git/GitHub.

PROJECTS

- **Agentic RAG Knowledge Assistant** — Apr 2026 – Aug 2026
  AI-Powered Document Q&A Platform | Individual Project
  - Architected a full-stack Retrieval-Augmented Generation (RAG) platform supporting PDF, DOCX, TXT, and Markdown uploads, enabling cited answers across 100+ documents with high retrieval accuracy.
  - Designed a production RAG pipeline using chunking, OpenAI text-embedding-3-large, Qdrant, retrieval, and citation generation, achieving 97% source-grounded responses and reducing unsupported answers.
  - Orchestrated multi-agent workflows with LangGraph using Planner, Retriever, Reasoning, and Response agents, decreasing average query latency to 10 seconds for multi-step requests.
  - Secured FastAPI APIs with JWT authentication and added multi-turn conversation memory using PostgreSQL, supporting 500+ concurrent sessions and improving session persistence by 30%.
  - Containerized the application stack with Docker Compose and configured GitHub Actions CI pipelines running Pytest cases, maintaining 95% automated test coverage.
  - Tech Stack: Next.js, TypeScript, FastAPI, LangGraph, Qdrant, OpenAI API, PostgreSQL, Docker
- **AI Interview Coach** — Dec 2025 – Mar 2026
  Intelligent Mock Interview Platform | Group Project
  - Created an AI-powered mock interview platform using OpenAI APIs, generating role-specific questions and evaluating responses in real time with 94% content relevance.
  - Engineered prompt-engineering workflows to simulate adaptive interviewer behavior, generating context-aware follow-up questions with 90% relevance.
  - Implemented a backend feedback and scoring engine, generating Json-formatted assessments across communication, accuracy, and response depth with 92% scoring consistency.
  - Constructed modular REST APIs using Node.js and Express for prompt generation, response processing, and session management, supporting concurrent interview sessions with 97% request success.
  - Integrated persistent session tracking with MongoDB, managing interview records and user progress across 1000+ active users.
  - Tech Stack: Node.js, Express.js, OpenAI API, EJS, MongoDB

SKILLS SUMMARY

- **Languages**: C, Java, Python, JavaScript
- **AI / LLM Engineering**: Retrieval-Augmented Generation (RAG), LangGraph, LangChain, Vector Databases (Qdrant), Semantic Search, OpenAI API, Multi-Agent Systems
- **Frontend**: React.js, Next.js, Redux, Tailwind CSS, HTML5, CSS3
- **Backend**: Node.js, Express.js, FastAPI, REST APIs, Passport.js, JWT Authentication
- **Databases**: MongoDB, MySQL, PostgreSQL
- **Tools**: Git, GitHub, Cloudinary, Mapbox, Docker
- **Coursework**: Data Structures & Algorithms, DBMS, Operating Systems, Machine Learning, AI, Computer Networks, Software Engineering-Agile, Scrum, Kanban, Blockchain and CryptoCurrency

EXTRA-CURRICULAR ACTIVITIES

Competitive Programming: Solved 300+ DSA problems across LeetCode and GeeksforGeeks using optimized algorithms and analysis.